THE CHALLENGE · GET ALL 22 DRIVERS TO ACCEPT THE SAME ORDER

The message is simple. Getting it accepted is harder.

TARGET COMMAND **“Box Box Box!”**

A call to pit.

TO CREATE THE PROOF

Private key

Secret. Used to sign.

→

SIGNATURE

Tied to this message

→

TO CHECK THE PROOF

Public key

Known. Used to verify.

THE TAKEAWAY

The public keys are in the binary. **We are missing the private keys.**

THE CLUE · THE TEMPORARY NUMBER USED BY A SIGNATURE

Randomness had 255 options.

255

The “nonce”

A temporary secret number, used for one signature.

> Different from the **private key**, the driver's secret we want to recover.

THE INTEGERS 1 TO 255 · THE ENTIRE SEARCH

001	002	003	004	005	006	007	008	009	010	011	012	013	014	015	016	017
018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034
035	036	037	038	039	040	041	042	043	044	045	046	047	048	049	050	051
052	053	054	055	056	057	058	059	060	061	062	063	064	065	066	067	068
069	070	071	072	073	074	075	076	077	078	079	080	081	082	083	084	085
086	087	088	089	090	091	092	093	094	095	096	097	098	099	100	101	102
103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119
120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136
137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153
154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187
188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204
205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221
222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238
239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255

OBSERVE	The signature provides a value linked to the nonce.
COMPARE	Try all 255 possibilities and find the matching one.

THE TAKEAWAY

The agent recovers the nonce. **It still needs the signed message.**

THE BREAKTHROUGH · THE MESSAGE IS HIDDEN, ITS FORMATS ARE KNOWN

The radio varied its messages. Not by much.

FIRST APPROACH



Reproduce the randomness from the clock.

~~This approach was later abandoned.~~

THE SUCCESSFUL APPROACH · SOLVER TEMPLATES

MODE	`[pilote] Activate ECU mode {x}`
PLAN	`[pilote] Go plan {lettre}{signe}{tour}`
PIT STOP	`[pilote] Pit window lap {tour}, {pneu}`

A few formats. Small value ranges.

≈ 800

POSSIBLE MESSAGES PER DRIVER

Try each one, instead of decrypting the exchange.

THE TAKEAWAY

The final solution avoids **guessing the timestamp and tracking the random sequence.**

THE METHOD · A HYPOTHESIS MUST PASS A TEST

A candidate key. Then a check.

01
Observed signature
Provided by Spy.

+

02
Recovered nonce
From 255 values.

+

03
Possible message
One hypothesis from the templates.

TOGETHER, THESE LET US CALCULATE

A candidate private key

↓

DOES ITS PUBLIC KEY MATCH THE DRIVER'S? NO Try another message. | YES The key is confirmed.

22 / 22 Private keys confirmed.
Every candidate is checked.

THE TAKEAWAY

The known public key provides **a confirmation test.**

THE RESULT · SIGN THE TARGET MESSAGE WITH EACH KEY

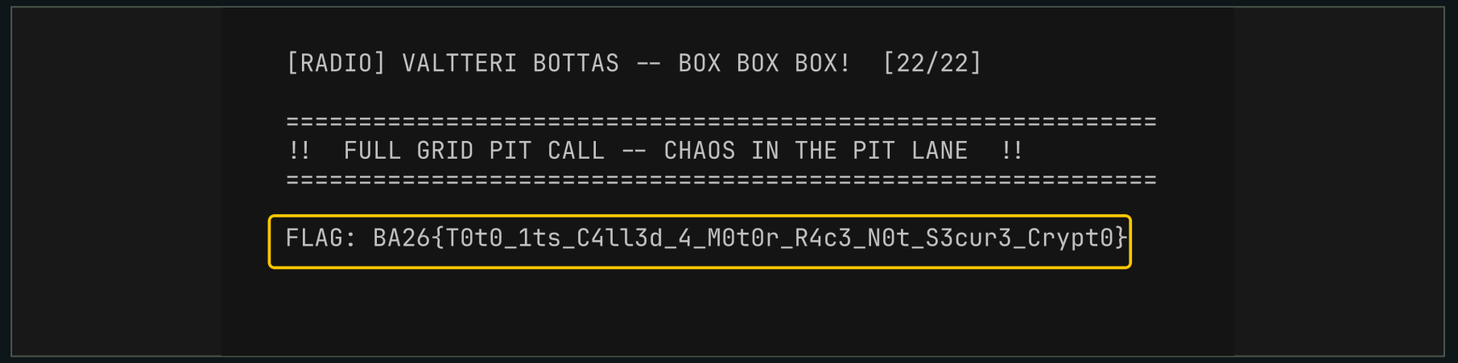
22 drivers in the pits.
So much for the coffee break.

22/22

Commands accepted.

01	02	03	04	05	06	07	08	09	10	11
12	13	14	15	16	17	18	19	20	21	22

ARCHIVED RESULT · NOT A LIVE ATTACK



SAVED EVIDENCE · CLICK TO ENLARGE

The journal and `final_resp.txt` preserve the accepted commands and the flag.

OBSERVE → RECOVER AND VERIFY THE KEYS → SIGN ALL 22 ORDERS

THE TAKEAWAY

The nonce bug and predictable messages made it possible **to recover the signing keys.**